

Question list

29 September 2026 11:22

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# =====  
#          PANDAS MINI PROJECT  
#          Employee Data Analysis  
#          Questions: 1–26  
# =====
```

```
# Q1 — Select Columns  
# Select the Name, Department, and Salary columns.
```

```
# Q2 — Conditional Filtering  
# Find employees whose:  
# - Salary is greater than 50,000  
# - Performance is greater than 85  
# Display Name, Salary, and Performance.
```

```
# Q3 — Using .isin()  
# Find employees who belong to either the IT or Sales department.  
# Display Name, Department, and Salary.
```

```
# Q4 — Using .between()  
# Find employees whose Age is between 25 and 32, inclusive.  
# Display Name, Age, and Department.
```

```
# Q5 — Multiple Conditions  
# Find employees who:  
# - Belong to IT or Sales  
# - Have Salary between 40,000 and 60,000  
# - Have Performance greater than 80  
# Display Name, Department, Salary, and Performance.
```

```
# Q6 — Create a New Column  
# Create a new column called Annual_Salary.  
# Calculate Annual_Salary = Salary × 12.
```

```
# Q7 — Performance Level  
# Create a Performance_Level column using np.where():  
# - Performance >= 90 → Excellent  
# - Performance >= 80 → Good  
# - Performance >= 70 → Average
```

- Otherwise → Needs Improvement

Q8 — Extract First Name

Create a First_Name column by extracting the first name
from the Name column.

Example: "Aman Kumar" → "Aman"

Q9 — Extract and Filter Last Name

Extract the last name from the Name column.

Find employees whose last name is Kumar.

Display Name, Department, and Salary.

Q10 — String Transformation

Create a Name_Upper column containing employee names
in uppercase.

Q11 — GroupBy

Calculate the average Salary for each Department.

Q12 — Multiple Aggregations

For each Department, calculate:

- Employee count

- Average Salary

- Maximum Salary

- Minimum Salary

- Average Performance

Q13 — Filter + GroupBy

Consider only employees whose Performance is greater than 80.

Calculate the average Salary for each Department.

Q14 — GroupBy + Sorting

Calculate the average Salary for each Department.

Sort the result from highest to lowest average Salary.

Q15 — Maximum Annual Salary

For each Department, find the maximum Annual_Salary.

Q16 — Filter + Multiple Aggregations

Consider employees who:

- Have Experience >= 3

- Have Performance >= 80

#

For each Department, calculate:

```
# - Employee count
# - Average Salary
# - Average Performance
# - Maximum Salary
#
# Sort by Average Salary in descending order.
```

```
# Q17 — Department Performance Report
# For each Department, calculate:
# - Employee count
# - Average Salary
# - Average Performance
# - Maximum Performance
# - Minimum Performance
#
# Sort by Average Performance in descending order.
```

```
# Q18 — High Performers
# Find employees who satisfy all three conditions:
# - Performance >= 85
# - Salary > 40,000
# - Experience >= 3
#
# Display:
# - Name
# - Department
# - Salary
# - Experience
# - Performance
#
# Sort by Performance in descending order.
```

```
# Q19 — Missing Values
# Find the number of missing values in every column
# of the employees DataFrame.
```

```
# Q20 — Handle Missing Age
# Fill missing values in the Age column using the median Age.
# Verify that no missing values remain in Age.
```

```
# Q21 — Duplicate Check
# Check whether the employees DataFrame contains duplicate rows.
# Display the duplicate rows.
# Calculate the total number of duplicate rows.
```

```
# Q22 — Merge / SQL INNER JOIN
# Merge the employees and departments DataFrames using
```

the Department column.

Use an INNER JOIN.

#

Display:

- Name

- Department

- Manager

- Office

Q23 — Extract Manager's Last Name

Create a Manager_Last_Name column from the Manager column.

Example: "Mr. Verma" → "Verma"

Q24 — Filter After Merge

Using the merged DataFrame, find employees who:

- Belong to IT or Sales

- Have Performance > 85

#

Display:

- Name

- Department

- Manager

- Performance

Q25 — Complete Pandas Analysis

Starting with the employees DataFrame:

#

1. Create Annual_Salary.

2. Create Performance_Level.

3. Create First_Name.

4. Merge employees with departments.

5. Keep only employees who:

- Belong to IT or Sales

- Have Performance >= 80

- Have Experience >= 3

6. Display:

- First_Name

- Department

- Manager

- Annual_Salary

- Experience

- Performance

- Performance_Level

7. Sort by Performance in descending order.

Q26 — Final Department Analytics

Create a department-level report containing:

- Department

- Employees

```
# - Avg_Salary
# - Avg_Experience
# - Avg_Performance
# - Max_Salary
#
# Sort by Avg_Performance in descending order.
```

```
# =====
#          END OF QUESTIONS
# =====
```